WEEK 7 · LECTURE L1 · UNIT III

People Analytics: The Analytics Ladder

From describing what happened to recommending what to do, and why most HR functions stand on the lower rungs.

R K Singh · Senior Professor, Department of Commerce · Delhi School of Economics, University of Delhi

DSEH001 · Digital Transformation in Human Resource Systems · M.Com. PGCF 2025 · Exam preparation edition · Updated 25 October 2026

AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Distinguish descriptive, diagnostic, predictive and prescriptive analytics, and explain why each rung rests on the ones below.
2. Define HR analytics (Marler and Boudreau) and state what the evidence shows and the conditions for impact.
3. Locate an organization on Davenport, Harris and Shapiro’s six uses, and start from a business problem (Fitz-enz and Mattox; Rasmussen and Ulrich).
4. Distinguish explanation from prediction (Shmueli), leading from lagging indicators, and correlation from cause.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 3 and CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

An HR head says: “We have no reliable headcount report yet, so let us skip ahead and buy a model that predicts who will resign.” What is the strongest objection?

A

Prediction models are too expensive for HR

B

Each rung of analytics rests on the ones below; a prediction built on untrusted facts produces confident errors

C

Resignations cannot be predicted at all

D

Predictive models are illegal under the DPDP Act

EXAM LENS

The answer is B. An organization that cannot count its exits reliably should not be predicting them.

THE ANALYTICS LADDER

Four kinds of question

Rung	Question	HR example	Needs
1 · Descriptive	What happened? Hindsight	Exits last year, by unit	Clean, integrated data and agreed definitions
2 · Diagnostic	Why did it happen? Insight	Do staff under weakly rated managers leave more?	Statistical reasoning about comparisons and confounders
3 · Predictive	What is likely to happen? Foresight	Who is likely to leave next quarter?	Enough history, and a fairness check
4 · Prescriptive	What should we do?	Which managers to coach first	Causal understanding, ideally tested

EXAM LENS

Each rung depends on the ones below, and a higher rung is not automatically more valuable: a descriptive view a store manager uses weekly beats an unused prediction. This ladder grades the question; Bersin’s (Week 2) grades the organization.

After Gartner’s analytic ascendancy model and Davenport and Harris (2007).

MARLER AND BOUDREAU (2017)

What counts as HR analytics, and what the evidence shows

Definition

An HR practice enabled by information technology that uses descriptive, visual and statistical analysis of data on HR processes, human capital, organizational performance and external benchmarks to establish business impact and enable data driven decisions.

What the review found

Few peer reviewed empirical studies; adoption lags the hype; evidence of impact limited but promising where it exists. Absence of evidence is not evidence of no effect.

Conditions for impact

Link HR data to business outcomes rather than reporting HR activity; build HR's own analytical capability; build results into decision processes.

EXAM LENS

Analysis that never reaches a decision is not analytics, on this definition. Quote its key elements.

DAVENPORT, HARRIS AND SHAPIRO (2010)

Six uses of talent analytics, simple to advanced

1 · Human capital facts
One trusted set of workforce indicators.

2 · Analytical HR
Segmenting data to see where HR practice works.

3 · Investment analysis
Which HR actions affect business outcomes.

4 · Workforce forecasts
Anticipating shortages and surpluses.

5 · Talent value models
Why employees stay or leave.

6 · Talent supply chain
Staffing to demand in near real time.

EXAM LENS

Leading firms treat talent decisions with the rigor of other investments. The cases are large, data rich firms such as Google and Harrah’s; most Indian HR functions are honestly at the first use.

FITZ-ENZ AND MATTOX (2014), CHAPTERS 1 AND 2

Start from the business question

Three levels

Descriptive (metrics, dashboards, scorecards), predictive (probability of future outcomes) and prescriptive (decision options and their likely business effects). Value grows upward; each level depends on the one below.

Five steps

Organize the data (often the hardest part), display it, relate it to other data (careful benchmarking), model the future you want, evaluate whether the change happened.

Getting started

Begin with a problem senior management cares about, not a favorite metric; tie a 2% fall in attrition to cost or revenue; turnover is the usual first project; enlist a sponsor outside HR.

EXAM LENS

Reporting activity counts what HR did; explaining outcomes links people data to results. The value lies in the second.

THEORY IN DEPTH · RASMUSSEN AND ULRICH (2015)

Analytics should start with a business problem

The core idea

HR analytics risks becoming a management fad if it starts from available data rather than decisions.

How it works

Begin with a business challenge, then find the data and method that answer it.

Limits and critique

Some discovery does come from exploring data; the warning is against exploration with no purpose.

IN PRACTICE

Start from “why are frontline sites losing people?”, not from “what can our dashboard show?”

EXAM LENS

Use this to critique dashboards first analytics programs.

THEORY IN DEPTH · SHMUELI (2010)

To explain or to predict: two different goals

The core idea

Models built to explain causes and models built to predict outcomes are judged differently.

How it works

Prediction rewards accuracy on new data; explanation needs variables that truly cause the outcome.

Limits and critique

Many studies blur the two and claim causes from predictive accuracy.

IN PRACTICE

A flight risk model can predict well and still give managers no idea what to change; a variable such as age can be a proxy, not a driver.

EXAM LENS

Distinguishing explanation from prediction is a mark of analytical rigor.

CORRELATION, CAUSE AND TIMING

Three traps, and leading against lagging

Confounding

A third factor drives both: tenure may explain both pay and loyalty.

Reverse causation

Engaged people may seek training, rather than training creating engagement.

Selection

Only those who stayed are measured, so survivors flatter the program.

Lagging indicators

Record the outcome after it happens: attrition counts people already gone.

Leading indicators

Move before the outcome: engagement and manager quality often shift before exits rise. Show the leading indicator really comes first (temporality) before relying on it.

EXAM LENS

The prescriptive rung needs causal evidence: experiments, or designs that rule out the three traps. A dashboard built only on lagging indicators reports the loss after it can be prevented.

FROM QUESTION TO ANSWERABLE QUESTION

Five parts, and two kinds of trap

Make it answerable

Population (who), outcome (what counts), comparison (against what), data (which tables, joined how) and the decision it informs.

Data traps

Split records, inconsistent definitions, missing populations (frontline in surveys, gig workers in HR data), data entered for administration, not analysis.

Capability traps

No owner for analytics; staff who can run reports but not frame questions; analysis handed to vendors who do not know the business; leaders who ask for numbers but not decisions.

Course framing.

APPLY

A resignation model before a headcount report

Teaching case (Course framing)

An Ahmedabad pharmaceutical firm with 9,000 staff has three headcount figures that disagree. A vendor offers a resignation prediction model with 90% claimed accuracy. The CHRO wants it live in eight weeks to impress the Board.

Work it through

1. On which rung is the request, and which rungs are missing?
2. Which of the six uses must come first, and why?
3. What would Marler and Boudreau ask before buying?
4. What does 90% accuracy hide if only 8% of staff resign a year?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
A higher rung is always better	Each rung rests on the ones below; a used description beats an unused prediction
Any analysis of HR data is analytics	Marler and Boudreau require a link to business impact and to decisions
A strong correlation tells us what to do	Prescription needs causal evidence
A model that predicts well explains why	Shmueli: prediction and explanation are different goals

COMPARE AND EVALUATE

Six lenses on people analytics

Lens	Use it to	Watch out for
Analytics ladder	Place the question	Higher is not automatically better
Marler and Boudreau (2017)	Define analytics and judge evidence	Published evidence lags practice
Davenport et al. (2010)	Locate maturity of use	Cases from data rich giants
Fitz-enz and Mattox (2014)	Start and sequence a project	Practitioner text
Rasmussen and Ulrich (2015)	Avoid the fad	Some discovery is exploratory
Shmueli (2010)	Separate explanation from prediction	Often blurred in practice

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

A Bengaluru IT firm wants to move straight to predictive attrition analytics. Using the analytics ladder, Marler and Boudreau’s review and Davenport, Harris and Shapiro’s six uses, advise the CHRO on sequencing, and note one risk of prediction. **20 marks**

A strong answer: explains the four rungs and their dependency; applies Marler and Boudreau’s definition and conditions; uses the six uses to show what must exist first; names a risk such as accuracy masking rare events or bias in the history.

“Most HR functions are trying to predict before they can describe.” Evaluate this claim using the analytics ladder and the evidence on HR analytics. **20 marks**

A strong answer: defines the rungs; uses Marler and Boudreau and Rasmussen and Ulrich; applies Shmueli; argues for sequencing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

Distinguish explanatory from predictive models, and discuss the implications for flight risk analytics in HR. **20 marks**

A strong answer: explains Shmueli's distinction; shows why accuracy does not imply actionable causes; addresses harms of false flags.

Distinguish leading from lagging indicators with HR examples. **10 marks**

A strong answer: gives correct examples and the temporality test.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Know which rung your question is on.**
   And whether the rungs beneath it are sound.

2. **Start from a business problem.**
   Analysis that reaches no decision is not analytics.

3. **Prediction is not explanation.**
   Prescription needs causal evidence.

4. **Leading indicators let you act before the loss.**
   Test that they really come first.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Marler, J. H., & Boudreau, J. W. (2017). An evidence based review of HR analytics. *IJHRM, 28*(1), 3 to 26.
- Davenport, T. H., Harris, J., & Shapiro, J. (2010). Competing on talent analytics. *HBR, 88*(10), 52 to 58.
- Fitz-enz, J., & Mattox, J. R., II (2014). *Predictive analytics for human resources*, Chapters 1 and 2.

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Davenport, T. H., & Harris, J. G. (2007). *Competing on analytics*. Harvard Business School Press.
- Davenport, T. H., Harris, J., & Shapiro, J. (2010). Competing on talent analytics. *Harvard Business Review, 88*(10), 52 to 58.
- Fitz-enz, J., & Mattox, J. R., II (2014). *Predictive analytics for human resources*. Wiley.
- Marler, J. H., & Boudreau, J. W. (2017). An evidence based review of HR analytics. *The International Journal of Human Resource Management, 28*(1), 3 to 26. https://doi.org/10.1080/09585192.2016.1244699
- Rasmussen, T., & Ulrich, D. (2015). Learning from practice: How HR analytics avoids being a management fad. *Organizational Dynamics, 44*(3), 236 to 242. https://doi.org/10.1016/j.orgdyn.2015.05.008
- Shmueli, G. (2010). To explain or to predict? *Statistical Science, 25*(3), 289 to 310. https://doi.org/10.1214/10-STS330